
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shifts introduce uncertainty that undermines the robustness and gener-
2 alization capabilities of machine learning models. Conventional wisdom holds that
3 learning causal-invariant representations improves robustness to such shifts, yet a
4 growing body of empirical evidence points consistently in the opposite direction:
5 (i) across a wide range of real-world benchmarks, empirical risk minimization
6 (ERM) matches and frequently surpasses state-of-the-art out-of-distribution (OOD)
7 generalization methods, and (ii) its OOD performance improves further when
8 *all* available covariates—not just the causal ones—are used. These two patterns
9 recur across datasets rather than appearing in isolated cases. Drawing on both
10 empirical and theoretical evidence, we trace this reproducible advantage to hidden
11 confounding. Shifts in hidden confounders induce distribution changes that vio-
12 late the assumptions underlying existing OOD generalization approaches. Under
13 exactly these conditions, we prove that effective generalization requires learning
14 environment-specific relationships rather than relying on invariant ones alone. We
15 further show that models augmented with proxies for hidden confounders measur-
16 ably reduce the difficulty posed by hidden confounding shifts. Consistent with this
17 account, our experiments reveal that the ERM-based XGBoost attains the highest
18 average accuracy on both in-distribution and OOD test data and the strongest
19 alignment between our predictive-information decomposition and observed accu-
20 racy gains, outperforming the invariance- and robustness-based baselines by wide
21 margins. Together, these findings offer new theoretical insight and concrete, empir-
22 ically grounded guidance for designing robust OOD generalization algorithms and
23 principled covariate selection strategies.

1 Introduction

24 Generalization—the ability to draw reliable conclusions about unseen data based on observed data—is
25 central to numerous scientific fields. In medicine and the social sciences, it is embodied as *external*
26 *validity*, ensuring findings from one population are applicable to a different population [13]; in ecology,
27 it supports *space-for-time substitutions*, where spatial variation is used as a proxy for temporal change
28 to infer long-term ecological patterns [60]; and in engineering, it drives *robust control*, where models
29 must maintain performance in the presence of unmodeled disturbances [35]. In recent years, machine
30 learning has become a powerful tool for learning generalizable models [100, 88, 49].

32 To generalize well, machine learning models must be robust to distribution shifts between training and
33 test data. For example, a model trained to predict *food stamp reciprocity* based on *household attributes*
34 in one state should be capable of adapting and performing well when deployed in another state. A
35 model is said to achieve *out-of-distribution (OOD) generalization* when it maintains its performance
36 on both in-distribution (ID) data (from which training data are sampled) and out-of-distribution
37 (OOD) test data. Over the last decade, various families of methods such as domain generalization [56,
38 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94]
39 have been proposed to achieve provable OOD generalization under specific assumptions. However,
40 under careful model selection [30, 38, 87], models based on standard *empirical risk minimization*
41 (*ERM*) [86] do not merely trail the specialized methods—they repeatedly match or exceed them
42 in OOD generalization across a broad spectrum of real-world applications [30, 24, 50, 57, 67, 87].

Reinforcing this pattern, a recent empirical study [57] spanning 16 real-world tabular datasets reports a consistent, dataset-after-dataset gain: incorporating *all* available covariates when predicting the outcome—regardless of whether they directly affect it—improves OOD generalization rather than degrading it. The fact that these two effects hold so uniformly, and in direct tension with prevailing assumptions, motivates a deeper investigation into the mechanisms underlying OOD generalization.

Distribution shifts are commonly observed when data originate from different environments [71]. For example, *food stamp reciprocity* may differ across states, because each state operates under different eligibility rules, leading to distinct, environment-specific distributions. In such data, certain statistical relationships between covariates and the outcome may stay consistent across all environments, referred to as *invariant relationships* [6]. Identifying and learning such invariant relationships ensures the development of models with OOD generalization [6, 59, 31, 66, 56, 98, 63]. A special kind of invariance is known as *causal invariance*, where the causal relationships in the data stay invariant across environments [71, 59, 6]. However, in many real-world settings, not all relevant covariates required for predicting an outcome are observed, due to limitations in data collection, privacy constraints, or measurement errors [14, 52, 20, 48]. Their absence disrupts the invariant relationships needed for models to achieve generalization. This issue is further compounded when the unobserved variables are confounders that influence both the observed covariates and the outcome. In practice, such hidden confounders are pervasive, and shifts in their distributions correspond to distinct environments. Ignoring these shifts not only undermines generalization performance [40, 4, 85, 62], but can also lead to learning incorrect relationships between observed covariates and the outcome.

Despite its importance, the impact of hidden confounding shifts on OOD generalization remains underexplored [4, 40, 85, 62]. This work offers both theoretical insights and empirical justifications that explain recent findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We motivate the need for studying confounding shift in OOD generalization from a causal perspective (§ 3) and explain why adding informative, non-causal covariates can improve performance.
- Specifically, we show that maximizing predictive information between model predictions and true outcomes demands explicitly learning environment-specific relationships. Consequently, invariant representations alone, while necessary, are insufficient (§ 4.1, § 4.2).
- We demonstrate that variables informative of either outcome or hidden confounders help in improving predictive information between model predictions and true outcomes. This underscores the importance of principled covariate selection in the presence of hidden confounding shift (§ 4.3).
- Our experiments on eight real-world tabular benchmarks and controlled synthetic data deliver three mutually reinforcing empirical findings (§ 5): (i) hidden confounding shift is pervasive—every one of the four measured shift types is statistically significantly nonzero across all eight datasets (p-value ≈ 0); (ii) the difference *conditional informativeness* – *residual* is positively correlated with both ID and OOD test accuracy, with the ERM-based XGBoost simultaneously achieving the largest value of this difference and the best accuracy (85.22% ID, 77.00% OOD on average); and (iii) adding informative, non-causal covariates consistently improves generalization, moving accuracy upward as covariate sets grow from causal to arguably causal to all covariates.

Overall, our work does more than explain the surprising and repeated effectiveness of ERM under real-world distribution shifts: it quantifies the subtle yet inevitable role of hidden confounding in OOD generalization and, in doing so, pinpoints exactly where more effective methods are needed.

2 Related work

Out-of-distribution (OOD) generalization encompasses various facets, with notable examples including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation assumes access to unlabeled data from the test set, whereas domain generalization relies on environment labels during training. Federated learning adopts a collaborative learning framework, tackling constraints such as communication efficiency and privacy when data originates from multiple environments. OOD detection focuses on identifying samples that differ significantly from the training distribution. A common goal of many of these methods is to learn invariant relationships. However, recent work suggests that additional inductive biases beyond invariance are required for improved generalization [46, 72, 93]. Ye et al. [93] argue that invariance of features is necessary but not sufficient for generalization and discuss the importance of *informativeness* of features for generalization. We also study the role of both invariance and informativeness for OOD generalization.

Proxy variable adjustment: When confounding variables are observed during training and unobserved at test time, Landeiro and Culotta [40] propose adjusting for confounding shifts to improve classifier performance. Building on proxy-based adjustment methods for causal effect estimation [55, 39], recent domain adaptation methods rephrase the problem of unknown distribution shifts as a causal effect identification problem [4, 85]. Recently, OOD generalization under hidden confounding shift has been considered under the assumption of overlapping confounder support [62]. In contrast to these approaches, we *explain how* proxy variables enhance OOD generalization.

All variable models vs causal models. Recently, Nastl and Hardt [57] introduced a benchmark study in which covariates are categorized into four groups: causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates. Their central empirical result is strong and uniform: across all 16 benchmark datasets, models trained on *all* covariates Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data—that is, the all-covariate models are never worse and are typically better on both axes at once. Theoretical explanations for this consistent dominance have been scarce, and we supply the missing account by identifying the scenarios and mechanisms that reproduce it. For linear causal models, anchor regression [68] introduces a framework that interpolates between two estimation paradigms: models that include all observed covariates and models that focus solely on causal covariates. Our aim is to explain *why* adding covariates that are not necessarily causal helps under a hidden confounding shift. Eastwood et al. [21] show that unstable covariates can boost performance when they carry information about the label, provided they are conditionally independent of the stable covariates given the label, and they adjust for the shift by inspecting the test domain without labels. Yet even against this tailored method, when applied to a medical real-world dataset not constructed for this particular problem [8], plain ERM remains competitive—mirroring the same ERM-is-hard-to-beat pattern reported by Nastl and Hardt [57]. This is consistent with the broader result that, under well-specified covariate shifts, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. Yet, real-world settings are rarely well-specified due to hidden confounding shift, which is the main focus of this work.

3 Manifestations of hidden confounding shift

In this section, we provide background on hidden confounding shift and motivate the need to address it using an example involving causal effect identification of observed covariates on the outcome.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , one may observe several distribution shifts between two environments e and e' as shown in Table 1. These distribution shifts usually result from a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes either $\mathbf{X}$ or Y or both (see Figure 1). The shifts in $\mathbb{P}(U)$ lead to shifts in observed distributions involving only $\mathbf{X}$ and Y . For instance, we can write: $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$. Thus,

when $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, we observe $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Similar arguments can be made about the distribution shifts of $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

Several existing methods for OOD generalization assume certain causal relationships among U , $\mathbf{X}$, Y that guarantee specific invariances. For instance, when $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, but conditional

covariate distribution stays invariant i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X}|Y, U) = \mathbb{P}(\mathbf{X}|Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, but the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ stays invariant i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These invariances may not hold in many scenarios because U usually causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this case, when $\mathbb{P}(U)$ shifts between environments, label shift, covariate shift, conditional covariate shift, and concept shift can all occur simultaneously. Neverthe-

Table 1: A summary of different distribution shifts.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

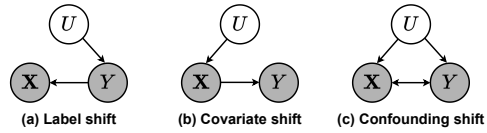


Figure 1: Causal graphs underlying distribution shifts.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

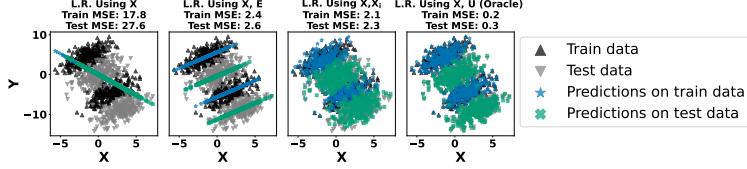


Figure 2: Head-to-head comparison of four linear regression (L.R.) models under hidden confounding shift with minimal environment overlap (i.e., distant μ_e); the four models are identical except for the information they receive. (i) Trained only on X , the model recovers a relationship with Y of the *wrong sign*, an instance of Simpson’s paradox. (ii) Adding environment-specific summary statistics of X (denoted E) flips this to the correct direction, though limited representational power caps the gain. (iii) Adding an informative covariate X_i that proxies for U yields a further reduction in error and the best OOD generalization among the non-oracle models. (iv) The oracle, trained on X and U , marks the best attainable performance; model (iii) comes closest to it.

less, confounding shift induces an invariance: $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ referred to as *stable confounding shift* [85, 4]. Since U is unobserved, this invariance is absent from the observed data.

Information-theoretic measures: Following Federici et al. [22], we use mutual information to quantify distribution shifts. Specifically, we use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$ where E is the environment variable to measure the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ respectively across the environments. As discussed earlier, existing methods for OOD generalization often assume either $I(\mathbf{X}; E | Y) = 0$ through the label shift assumption or $I(Y; E | \mathbf{X}) = 0$ through the covariate shift assumption. However, as discussed earlier, both of them may be nonzero under hidden confounding shifts.

Challenges with hidden confounding shift: Figure 2 makes the challenge concrete through a controlled comparison of four linear regression models that differ only in the information they are given. The outcome Y depends on X and a hidden confounder U , and environment-specific shifts in the mean of U , denoted μ_e , define distinct environments. The first model, trained solely on the observed covariate X , recovers a relationship whose sign is *opposite* to the truth: Y rises with X within every environment, yet the fitted trend falls—a textbook instance of Simpson’s paradox [73]. Crucially, the dedicated OOD generalization methods IRM [6], VREX [38], and Group DRO [69] exhibit the same failure, so invariance and robustness objectives offer no advantage here. The second model, supplied with environment-specific summary statistics of X (mean, standard deviation, and quantiles), reverses this outcome and recovers the correct X – Y direction, echoing the *backdoor adjustment* criterion [58] in which environment-specific information acts as a proxy for the confounder; we prove in § 4 that this environment-specific strategy is in fact optimal. This second model still leaves accuracy on the table because the environment statistics have limited representational capacity. The third model closes much of that gap: adding an informative covariate that proxies for U (§ 4.3) yields a further, visible reduction in error. As an upper reference, the fourth model—an oracle trained on both X and U —attains the best-possible mean squared error, and the third model approaches it far more closely than the first.

4 Impact of hidden confounding shifts on OOD generalization

In this section, we theoretically study how hidden confounding shifts impact various aspects of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the

194 pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$
 195 where $\mathcal{D}^e$ denotes observed data in an environment e .

196 To understand how hidden confounding shift impacts traditional objective functions, we explore
 197 different aspects of maximizing predictive information: $I(Y; \hat{Y})$ where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. $I(Y; \hat{Y})$
 198 quantifies how informative the prediction $\hat{Y}$ is about the true label Y , making it a natural and
 199 meaningful objective for many machine learning tasks. We start by defining two key properties of
 200 $\phi(\mathbf{X})$ —*informativeness* and *invariance*—both of which are crucial for OOD generalization [93].

201 **Definition 4.1** (Informativeness and Conditional Informativeness). *The informativeness of fea-*
 202 *tures $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informa-*
 203 *tiveness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as*
 204 *$\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

205 Conditional informativeness measures the information $\phi(\mathbf{X})$ provides about Y within each envi-
 206 ronment. Minimizing CINF implies information loss, and maximizing CINF can be undesirable in
 207 applications such as algorithmic fairness, as $\phi(\mathbf{X})$ may exploit sensitive or biased information within
 208 environments, leading to unfair predictions. While such biased representations can be avoided by
 209 minimizing $I(\phi(\mathbf{X}); E)$, it may reduce predictive performance, as we show in § 4.2. Considering the
 210 setting where no sensitive information is associated with E , we adopt the perspective that maximizing
 211 CINF can improve generalization performance, a view we follow and substantiate in § 4.1.

212 **Definition 4.2** (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$*
 213 *across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the*
 214 *invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e.,*
 215 *$\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

216 Invariance quantifies how consistent the representation $\phi(\mathbf{X})$ is across different environments for
 217 a given label Y . Minimizing invariance can lead to overfitting by preserving environment-specific
 218 information in $\phi(\mathbf{X})$, whereas maximizing invariance helps eliminate environment-specific dependen-
 219 cies, promoting invariant learning. Extending the information-theoretic measures from § 3, we use
 220 $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift* respectively. Depending
 221 on the context, we use the term *concept shift* to denote either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

222 4.1 A general decomposition of predictive information

223 We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing
 224 random variables, and directed edges indicating direct causal influences between nodes. A *path*
 225 between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path*
 226 from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow$
 227 $X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an
 228 *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a
 229 *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and
 230 forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning
 231 on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are
 232 marginally independent but become conditionally dependent when conditioning on X_j or any of its
 233 descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by
 234 a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies
 235 in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$.
 236 If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

237 Now consider the predictive information $I(Y; \hat{Y})$, where the
 238 predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned represen-
 239 tations $\phi(\mathbf{X})$. We model the underlying causal relationships
 240 among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as shown in Figure 3. Here,
 241 U is a hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$,
 242 $\hat{Y}$ is redundant for reasoning about Y . That is, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$
 243 and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. E denotes an environment variable that
 244 captures shifts in $\mathbb{P}(U)$. That is, between any two environments,
 245 there is a shift in $\mathbb{P}(U)$. We first present a more general decom-
 246 position of predictive information without explicit consideration of how U influences $\mathbf{X}, Y$. A formal
 247 treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

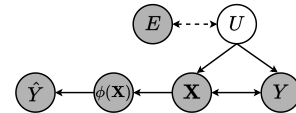


Figure 3: Bi-directed arrow between $\mathbf{X}$ and Y indicate that some covariates of $\mathbf{X}$ can cause Y , and some may be caused by Y .

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are presented in Appendix § A. Existing methods can be viewed as methods that explicitly minimize or maximize certain terms in Equation (1). For instance, IRM [6] enforces that a fixed classifier f remains the same across different environments. This is directly related to maximizing $-I(\phi(\mathbf{X}); E|Y)$ because if $\phi(\mathbf{X})$ were to carry extra environment-specific information conditional on Y , the fixed classifier would no longer be optimal in all environments. DANN [23] and the independence criterion for fair classification [22] aims to minimize environment-specific information in $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain properties of features are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may be good for fair classification, it may degrade predictive performance [98, 33, 22]. This can be seen from Equation (1), where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to obtain $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of the pair $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, its joint-distribution alignment objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is a constant, there exist methods that are robust to shifts in the distribution of labels [69, 95].

Recall that Equation (1) is derived without explicitly considering how U influences $\mathbf{X}, Y$. Knowing how U influences $\mathbf{X}$ and Y can further guide the effective handling of the terms in (1). That is, if $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, suggesting that minimizing $I(\phi(\mathbf{X}); E | Y)$ is a principled objective. Similarly, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, thereby motivating the minimization of $I(Y; E | \phi(\mathbf{X}))$. However, if $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, the interaction between the terms in Equation (1) becomes non-trivial, and it remains unclear what constitutes an ideal strategy for addressing them. We answer this question in the next section.

4.2 Predictive information under hidden confounding shift

To understand how hidden confounding variable U that cause both $\mathbf{X}$ and Y , impact the predictive information, we consider *two special cases* of the relationship between $\mathbf{X}$ and Y : (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, we obtain the inequalities in (2) because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y which results in additional information flow from $\phi(\mathbf{X})$ to E via Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

290 Proposition 4.2 treats the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but the first case dominates
 291 practice: 11 of the 16 datasets in recent benchmark studies [57, 50, 24] follow the causal structure
 292 $\mathbf{X} \rightarrow Y$, and even in the remaining 5 the covariates caused by Y are far outnumbered by those that
 293 cause Y . The result therefore applies, at least approximately, to essentially the entire benchmark
 294 suite. We state its key takeaway below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

295

296 **Maximizing conditional informativeness:** In a very recent work, Prashant et al. [62] proposed a
 297 mixture of experts (MoE) model for OOD generalization under hidden confounding shift, where
 298 each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on
 299 maximizing the performance within the environment corresponding to a particular value of the hidden
 300 confounder. Thus, learning MoE models aligns with the goal of maximizing the conditional infor-
 301 mativeness $I(\phi(\mathbf{X}); Y | E)$. Equation (4) provides theoretical justification for the use of MoE-type
 302 models. Beyond supporting such models, the Equation (4) also highlights the need for methods that
 303 operate under more general confounding shift settings. For instance, the method proposed by Prashant
 304 et al. [62] assumes overlapping confounding support i.e., $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, and
 305 the proxy variables to be discrete-valued.

The results in Figure 2 illustrate that, in a simple setting without confounding overlap and with a continuous-valued proxy, a linear regression model can successfully recover the correct causal relationship between observed covariates and the target across environments when provided with environment-specific information. This serves as a motivational example that emphasizes the need for OOD generalization methods that remain effective even when confounding support is limited and the confounder is continuous-valued.

306

307 **Minimizing residual:** The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information
 308 $\phi(\mathbf{X})$ provides about the true label Y beyond what is already contained in the prediction $\hat{Y}$. The
 309 residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional
 310 entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y})$
 311 $+ \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the *refinement error* that measures
 312 the model’s ability to distinguish between classes. Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus
 313 involves two complementary strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making
 314 $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can
 315 minimize the refinement error $H(Y | \hat{Y})$, for instance, through training techniques that minimize
 316 refinement error for accurate predictions before minimizing the calibration error [10].

317 4.3 Impact of additional informative variables

318 In § 4.2, we present the optimal strategies for generalization when the confounder is unobserved.
 319 In practice, we sometimes have access to proxies for the hidden confounding variable [4, 85, 62],
 320 which can be leveraged to substitute for the hidden confounding variable. We now study the impact
 321 of adding such non-causal but informative covariates on predictive information.

322 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 323 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 324 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

325 Since $U \rightarrow Y$, any covariate that is informative to U , is also informative to Y and vice-versa. From
 326 the predictive information decomposition in (1), to maximize $I(Y; \hat{Y})$, it is required to minimize the
 327 concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] suggest based on their empirical analysis that collecting
 328 additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ is more stable across environments i.e., reducing

concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative variables helps maximizing predictive information by maximizing or minimizing certain terms in (1). **Proposition 4.3.** *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

Proposition 4.3 pins down the direction of every affected term: adding informative covariates strictly increases conditional informativeness (5.i) and feature shift (5.ii) and strictly decreases concept shift (5.iii)—exactly the three movements that raise predictive information. The one countervailing effect is that variation also grows (5.iv), which may call for dedicated strategies to control it. Our experiments (Table 3, Figure 5) show these predicted directions holding in practice, with the beneficial terms dominating.

5 Experimental results

Our experiments on real-world and synthetic datasets are designed to answer three questions with quantitative evidence: (i) is hidden confounding shift actually present in real-world data, and how strong is it? (ii) which components of the decomposition in (1) track OOD generalization under hidden confounding shift, and how tightly? and (iii) does adding informative covariates improve generalization, and by how much? We evaluate on eight real-world tabular benchmark datasets—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *AS-SISTments*—with datasets and domain splits taken from the *TableShift* benchmark [24]. Performance is reported as ID test and OOD test accuracy, so every comparison below is on a common, directly interpretable scale. We benchmark five methods spanning three families: two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust learning method, Group DRO (GDRO) [69]. Each reported number is an average over 5 random hyperparameter choices (Appendix B); random seeds have a statistically insignificant effect on these datasets [24], so the differences we report reflect method behavior rather than seed variance. Additional setup details are in Appendix § B, and results under explicit hyperparameter tuning—which preserve the same rankings—are in Appendix C.

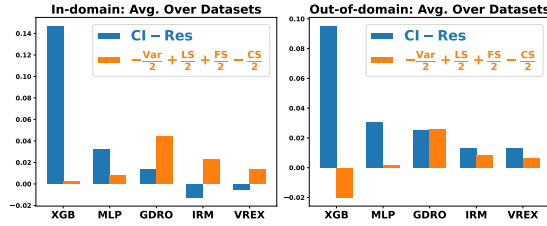
Hidden confounding shift: Recall that hidden confounding shifts induce observable shifts in the distributions $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} \mid Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y \mid \mathbf{X})$. Table 2 quantifies all four using mutual-information measures, and the evidence is unambiguous: conditional covariate shift, label shift, covariate shift, and concept shift are each significantly nonzero in *every one* of the eight datasets (p-value ≈ 0 against a zero mean), so no dataset escapes hidden confounding shift. The effect is far from marginal—concept shift, for instance, reaches 2.132 (Diabetes) and stays above 1.8 on seven of the eight datasets, and even the smaller shift terms are estimated tightly, with standard deviations at or below 0.006 throughout. To complement this quantitative signal, we query GPT-4o [1] for plausible hidden confounders across several benchmark datasets [24, 57] (Appendix § E); in the *food stamps* dataset, for example, unmeasured factors such as state-specific *economic policies* plausibly drive both *household* income and *food stamp* reciprocity.

Conditional informativeness vs. accuracy: Proposition 4.2 predicts that, under hidden confounding shift, predictive information is governed by the difference *conditional informativeness* – *residual*. Figure 4 tests this prediction on the first six datasets (those with more than one training environment [57, 24]) from Table 2, with per-dataset breakdowns in Appendix C. The empirical pattern matches the theory closely: this difference is positively correlated with both ID and OOD accuracy, and the method that maximizes it—XGB—is also the top performer, leading on average ID accuracy (85.22%) and OOD accuracy (77.00%) and topping the next-best model, MLP, by 1.58 and 3.71 percentage points respectively. The gap over the invariance/robustness methods is larger still: XGB exceeds GDRO, IRM, and VREX by up to 25.48 points ID and 13.94 points OOD. Consistent with Proposition 4.2, the orange bars in the per-dataset plots (Appendix C) collapse toward zero, because under hidden confounding shift the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes—leaving conditional informativeness minus residual as the dominant driver of accuracy. This ranking of methods reproduces the ID/OOD ordering previously reported in the literature [57, 24].

Informative covariates vs. accuracy: We next quantify how adding informative covariates improves OOD generalization. Following [57], we group all available covariates into three nested subsets,

Table 2: Every distribution-shift measure is significantly nonzero on every dataset, evidencing pervasive hidden confounding shift. Each cell reports mean $\pm$ standard deviation over 10 random subsets of 40,000 samples; one-sample t -tests against a zero mean ($\alpha = 0.05$) reject the null for all 32 dataset $\times$ measure combinations (p-value ≈ 0). Standard deviations are ≤ 0.006 throughout, indicating highly stable estimates.

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Higher *conditional informativeness* – *residual* tracks higher accuracy: the plotted difference is positively correlated with average ID and OOD test accuracy across the eight datasets. The table (right) shows the resulting ranking—XGB leads on both ID (85.22%) and OOD (77.00%), ahead of MLP (83.64%/73.29%), GDRO (78.40%/72.28%), IRM (71.17%/64.89%), and VREX (59.74%/63.06%)—so the model with the largest difference is also the most accurate.

causal $\subseteq$ arguably causal $\subseteq$ all covariates (Appendix § B), and measure how the terms in the decomposition (1) move as the covariate set grows. To make “moves in the right direction” precise, we define a *sign consistency metric*: when we enlarge the covariate set, a positively weighted term (e.g., conditional informativeness) should rise and a negatively weighted term (e.g., concept shift) should fall whenever accuracy improves; the metric is the fraction of such comparisons in which a term moves in its *beneficial* direction, so 1.0 means perfectly reliable alignment (see Appendix § B for the formal definition). Table 3 shows two clear patterns. First, reducing concept shift is reliably beneficial for *all* five methods (sign consistency ≥ 0.89 , up to 0.94 for GDRO). Second, and decisively, XGB attains the highest conditional-informativeness consistency (0.89, versus 0.64–0.78 for the other methods)—the very term Proposition 4.2 identifies as governing generalization—and this coincides with XGB posting the best accuracy in every covariate setting, rising from 81.24% (causal) to 84.84% (arguably causal) to 85.22% (all) on ID and from 69.92% to 76.72% to 77.00% on OOD. Growing the covariate set thus improves XGB’s OOD accuracy by 7.08 points overall. Dataset-specific results are in Appendix § C.

A caveat of real-world data is that we cannot verify whether the *all available covariates* set truly contains *every relevant or sufficiently informative* variable. We therefore corroborate the hypothesis on synthetic data with a fully known causal structure— $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ are informative covariates driven by the hidden confounder U —where every quantity is controlled. Here the predicted effect appears cleanly and monotonically (Figure 5): introducing $\mathbf{X}_I$ simultaneously raises conditional informativeness and feature shift while lowering concept shift and reducing MSE, exactly as (1) anticipates. The variation term stays pinned at zero throughout, so the improvement is attributable entirely to the beneficial terms. The full data-generating process and additional results are in Appendix § D.

6 Discussion, limitations, and future work

In summary, we decompose the predictive information between model outputs and true outcomes to expose the factors that limit OOD generalization (Equation (1)), and our experiments make that decomposition quantitatively actionable. Hidden confounding shift is not a corner case: it registers

Table 3: Sign consistency ($\uparrow$, higher is better) and accuracy across nested covariate sets. Concept shift (**CS**) is reliably beneficial for all five methods (≥ 0.89). The term that Proposition 4.2 identifies as decisive—conditional informativeness (**CI**)—is maximized by XGB (0.89 vs. 0.64–0.78 for the rest), and XGB correspondingly posts the best ID and OOD accuracy in *all three* covariate settings (highlighted). **Res**: residual, **Var**: variation, **FS**: feature shift, **C**: causal, **AC**: arguably causal, **A**: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

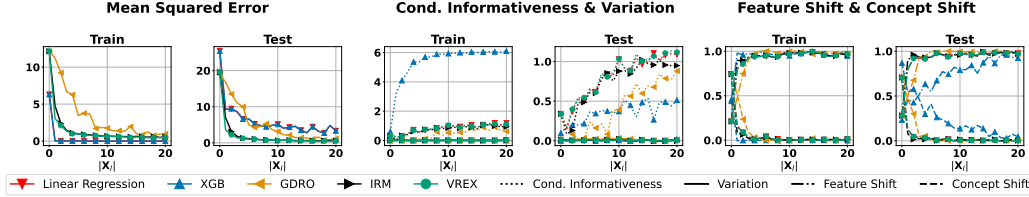


Figure 5: Controlled synthetic study: as more proxy variables $\mathbf{X}_I$ of U (informative to Y) are added, MSE decreases monotonically while conditional informativeness and feature shift rise and concept shift falls—all four quantities moving in the direction (1) predicts, with the variation term held at zero.

as significantly nonzero on all four shift measures across every one of the eight benchmark datasets (p-value ≈ 0), so no finite set of observed covariates or proxies can fully resolve the induced ambiguity. Because confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), treating these in isolation is ineffective—and the data bear this out, with the entangled **orange** terms collapsing toward zero and the difference *conditional informativeness* – *residual* emerging as the dominant, accuracy-tracking quantity (Equation (4)). Learning environment-specific input–output mappings is thus not merely preferable in principle but empirically superior: the method that best maximizes conditional informativeness, XGB, leads on both ID (85.22%) and OOD (77.00%) accuracy and in every covariate setting, while enlarging the covariate set lifts its OOD accuracy by 7.08 points. By pairing this evidence with theory, our work lays a foundation for more robust OOD methods. Key challenges ahead include: (i) developing algorithms effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without relying on untestable proxy assumptions, and (iv) modeling ambiguity from unobserved confounders to inspire new OOD-robust paradigms.

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Appendix

A Proofs of propositions

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proof. From the causal graph shown in Figure 3 of the main paper, in the causal substructure: $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$ or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the expansion instead of $\phi(\mathbf{X})$. $\square$

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

Proof. Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

721 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 722 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

723 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 724 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 725 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

726 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

727 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

728 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

729 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 730 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 731 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

732 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 733 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

734 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

735 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

736 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

737 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 738 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

739 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 740 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

741 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
742 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
743 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
744 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

745 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
746 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
747 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
748 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
749 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

750 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

751 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

752 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
753 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
754 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
755 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
756 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

757 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
758 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
759 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
760 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
761 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

762 B Experimental setup

763 This section details our experimental setup and the safeguards that make the reported comparisons
764 trustworthy. Random seeds have a statistically insignificant effect in many settings [24], so for
765 the main-paper results we report mean and standard deviation over five random hyperparameter
766 values (sampled from the ranges in Table B1) per dataset–method pair, ensuring the reported gaps
767 reflect method behavior rather than tuning luck. Mutual-information terms are computed with the
768 nonparametric entropy estimation toolbox [37, 77, 78], and, following [24], we evaluate them on
769 both in-distribution (ID) test data (from training domains) and out-of-distribution (OOD) test data
770 (from test domains) so that every measure is reported on the same two evaluation conditions used for
771 accuracy.

772 For consistency across datasets, we draw 20,000
773 random samples from each of the ID and OOD
774 test sets (a few datasets contain fewer). Sample
775 size has a negligible effect on the estimated
776 mutual-information terms: as Figure B1 illus-
777 trates, the terms in Equation (1) for the XGB
778 representations on the *income* dataset are es-
779 sentially flat as the sample size varies, so our
780 comparisons are stable to this choice. Features
781 $\phi(\mathbf{X})$ are taken from the layer preceding the
782 classification head; for XGBoost, we combine
783 the model’s output margins and SHAP values.
784 All experiments ran on a single NVIDIA RTX A6000 GPU.

785 For the experiments on informative covariates, we follow the categorization of Nastl and Hardt [57],
786 partitioning covariates into three nested sets—*causal* $\subseteq$ *arguably causal* $\subseteq$ *all covariates*—which
787 is exactly the axis along which our accuracy comparisons vary. Causal covariates have clearly
788 established influence on the target, with well-supported directionality and unlikely reverse causation,
789 though restricting to them risks omitting relevant parents due to knowledge gaps. Arguably
790 causal covariates have uncertain relationships, satisfying at least one of: (1) being known causal,

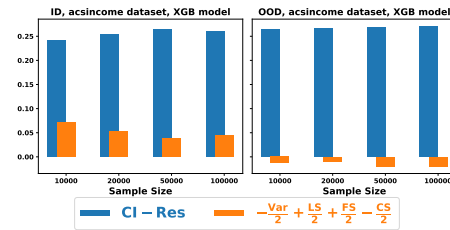


Figure B1: Robustness to sample size: on the *income* dataset, the evaluated predictive-information terms stay essentially flat as the sample size varies, confirming the reported measures are not artifacts of the chosen subset size.

Table B1: Hyperparameters and their possible values considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

791 (2) plausible but potentially bidirectional influence, or (3) a likely but unconfirmed causal effect;
 792 these approximate the true causal parents, though relationships may be confounded. Some datasets
 793 additionally contain *anti-causal* covariates that the target likely affects but not vice versa. The *all*
 794 *covariates* set includes every observed variable regardless of causal status. The sign consistency
 795 metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
 796 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
 797 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and test data. $\mu_s^{\text{tr}/\text{te}}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section reports per-dataset results that back the aggregate trends in the main paper. Figures C2, C3 break out the two groups of terms—*conditional informativeness – residual* (blue) and $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ (orange)—for each dataset (aggregated in main-paper Figure 4 Left). Two consistent patterns emerge. Per Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall performance on every dataset. More tellingly, the blue difference *conditional informativeness – residual* dominates the orange sum as a contributor to predictive information across datasets—a direct empirical fingerprint of hidden confounding shift, and concrete motivation for maximizing conditional informativeness to improve OOD generalization.

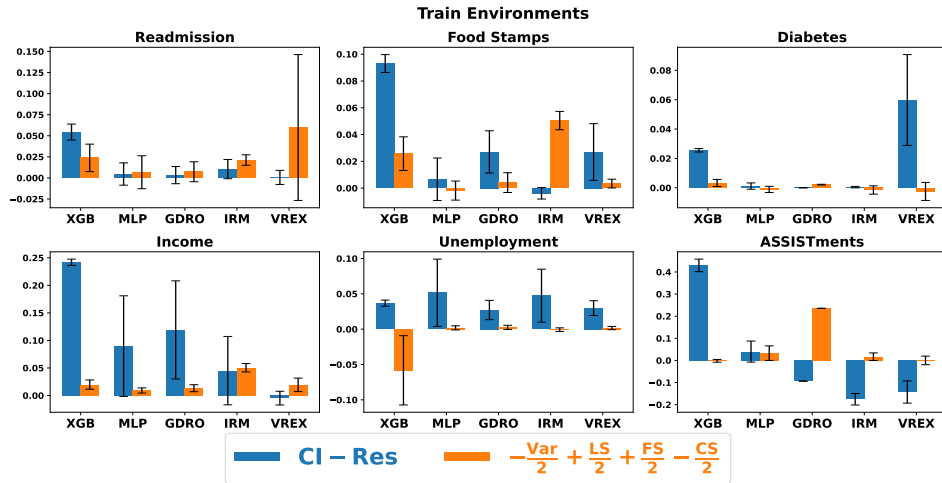


Figure C2: Per-dataset decomposition on ID data. On each dataset, the blue difference *conditional informativeness – residual* is the larger contributor to predictive information, while the orange sum stays near zero.

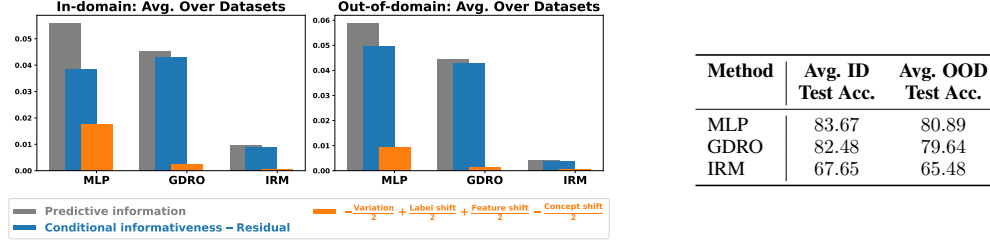


Figure C4: Results under explicit hyperparameter tuning (best ID-validation configuration) reproduce the main-paper trends. The difference *conditional informativeness* – *residual* remains positively correlated with average ID and OOD accuracy over the five datasets in the table (right), where MLP leads (83.67% ID / 80.89% OOD), followed by GDRO (82.48%/79.64%) and IRM (67.65%/65.48%). As before, the *blue* difference *conditional informativeness* – *residual* contributes more to predictive information than the *orange* sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

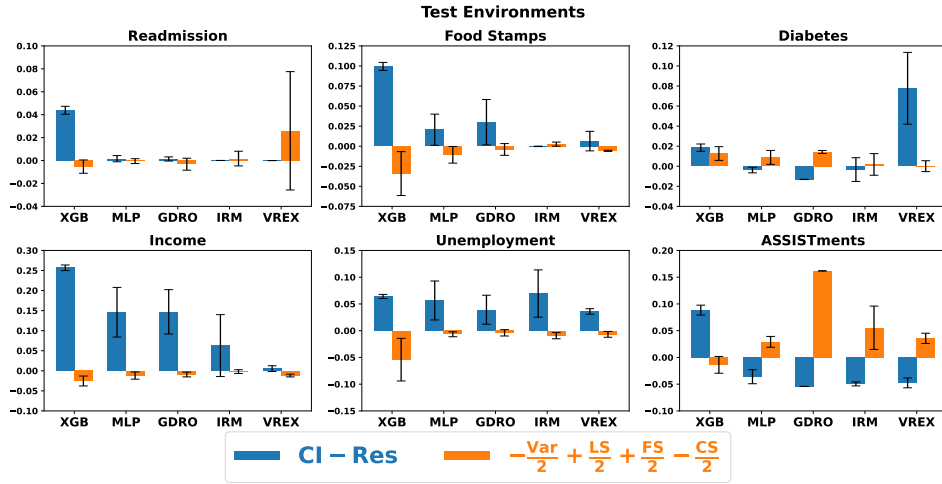


Figure C3: Per-dataset decomposition on OOD data. The same pattern as on ID data holds out of distribution: the *blue* difference *conditional informativeness* – *residual* dominates the near-zero *orange* sum on every dataset.

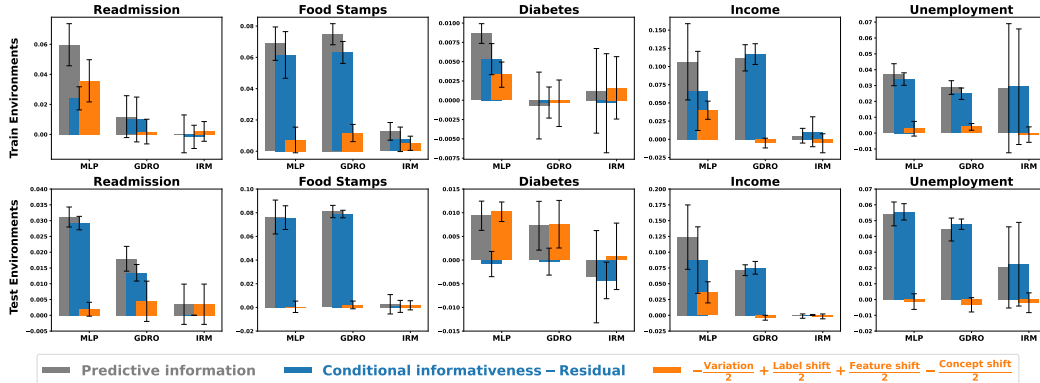


Figure C5: Per-dataset comparison of the predictive-information terms under hyperparameter tuning, showing the same dominance of the *conditional informativeness* – *residual* difference across datasets.

Table C2: Per-dataset ID and OOD test accuracy (mean $\pm$ std). MLP is the strongest of the three methods on most datasets, leading OOD accuracy on Readmission (61.75%), Food stamps (82.00%), Diabetes (83.22%), and Income (81.46%), and running essentially even with GDRO on Unemployment (96.04% vs. 96.10%). IRM trails throughout and exhibits far larger variance (e.g., ± 7.35 ID on Readmission and ± 36.5 on Diabetes), underscoring its instability relative to the ERM-based MLP.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 $\pm$ 0.14	61.75 $\pm$ 0.22	84.74 $\pm$ 0.03	82.00 $\pm$ 0.06	87.66 $\pm$ 0.02	83.22 $\pm$ 0.05
GDRO	60.47 $\pm$ 1.08	57.60 $\pm$ 0.42	84.47 $\pm$ 0.04	81.41 $\pm$ 0.09	87.47 $\pm$ 0.10	82.90 $\pm$ 0.10
IRM	50.28 $\pm$ 7.35	50.85 $\pm$ 1.74	80.91 $\pm$ 0.00	78.01 $\pm$ 0.00	42.54 $\pm$ 36.5	43.48 $\pm$ 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 $\pm$ 0.05	81.46 $\pm$ 0.01	97.27 $\pm$ 0.02	96.04 $\pm$ 0.05
GDRO	82.70 $\pm$ 0.05	80.16 $\pm$ 0.45	97.28 $\pm$ 0.01	96.10 $\pm$ 0.00
IRM	67.91 $\pm$ 0.00	60.20 $\pm$ 0.00	96.61 $\pm$ 0.00	94.84 $\pm$ 0.00

Figures C6–C15 give a fine-grained, per-method and per-dataset view of the individual terms in the decomposition (1), and they make the cancellation explicit. The second column (variation) and third column (feature shift) are close in magnitude on essentially every panel, so by (1) their difference nets out to near zero; the same near-cancellation holds between the fourth column (label shift) and fifth column (concept shift). As a result, in almost all settings the difference *conditional informativeness* - *residual* is the majority contributor to predictive information—a pattern that repeats across all five methods and every dataset, reinforcing the main-paper conclusion.

Hyperparameter tuning: To confirm the findings are not an artifact of random hyperparameter sampling, we additionally run a full grid search, selecting configurations by ID validation accuracy (not OOD)—a total of 1500 experiments (5 datasets $\times$ 3 models $\times$ 100 hyperparameter choices), each repeated over 5 random seeds for mean and standard deviation even though seed effects are typically negligible. The tuned results (Figure C4) tell the same story as the main paper: consistent with Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall performance, and the blue difference *conditional informativeness* - *residual* again contributes more to predictive information than the orange sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. The dataset-specific results in Figure C5 and Table C2 exhibit the same trends, so the conclusions are stable across both hyperparameter-selection protocols.

Potential limitations with predictive information decomposition analysis: Predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels and, by itself, does not certify accuracy. A stark example: a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation while scoring 0% accuracy, because mutual information quantifies *reduction in uncertainty* rather than correctness. Related caveats apply to the individual terms—conditional informativeness $I(\phi(X); Y|E)$ can mask harmful variation $I(\phi(X); E|Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ via spurious correlation. These edge cases do not arise for the reasonably performant models we study: across all eight datasets the decomposition tracks accuracy in the direction predicted by our theory, so it reliably distinguishes predictive information arising from generalizable patterns versus environment-specific artifacts—yielding an interpretable, quantitatively faithful diagnostic for targeted improvement.

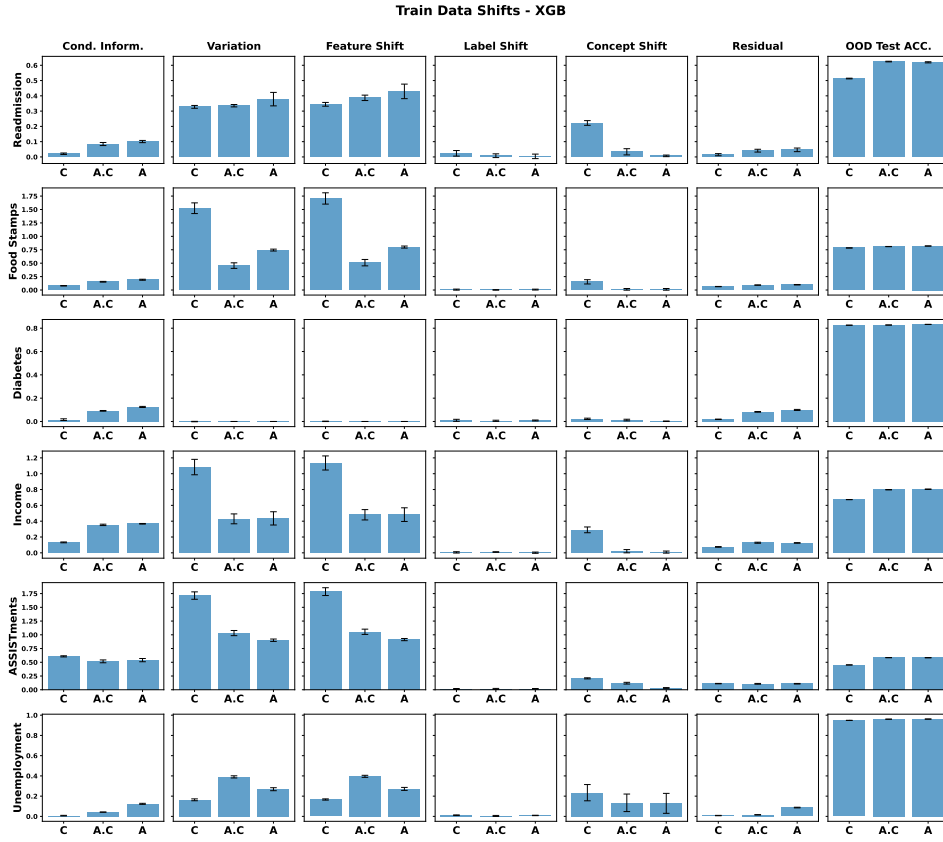


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

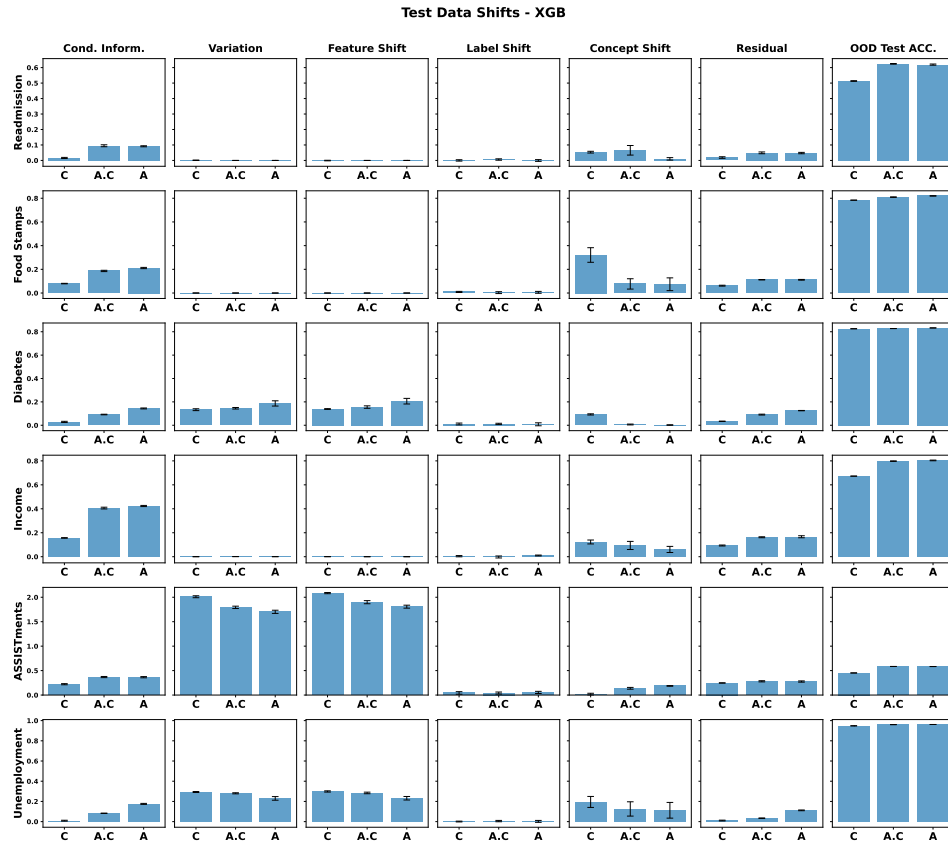


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

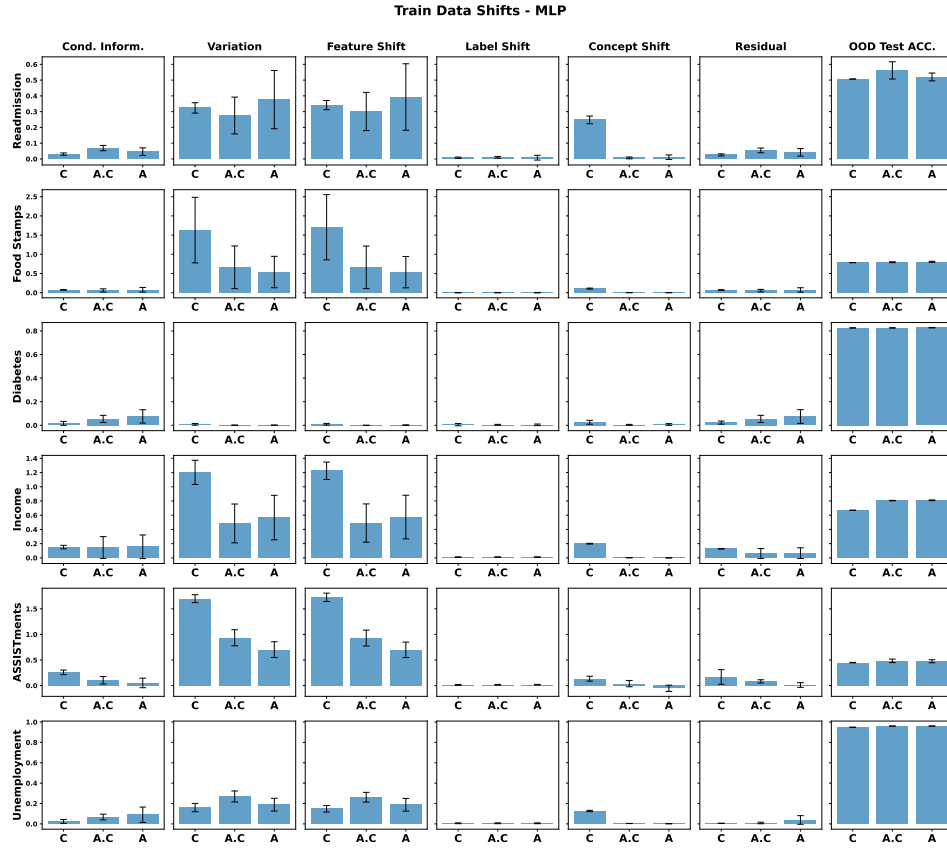


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

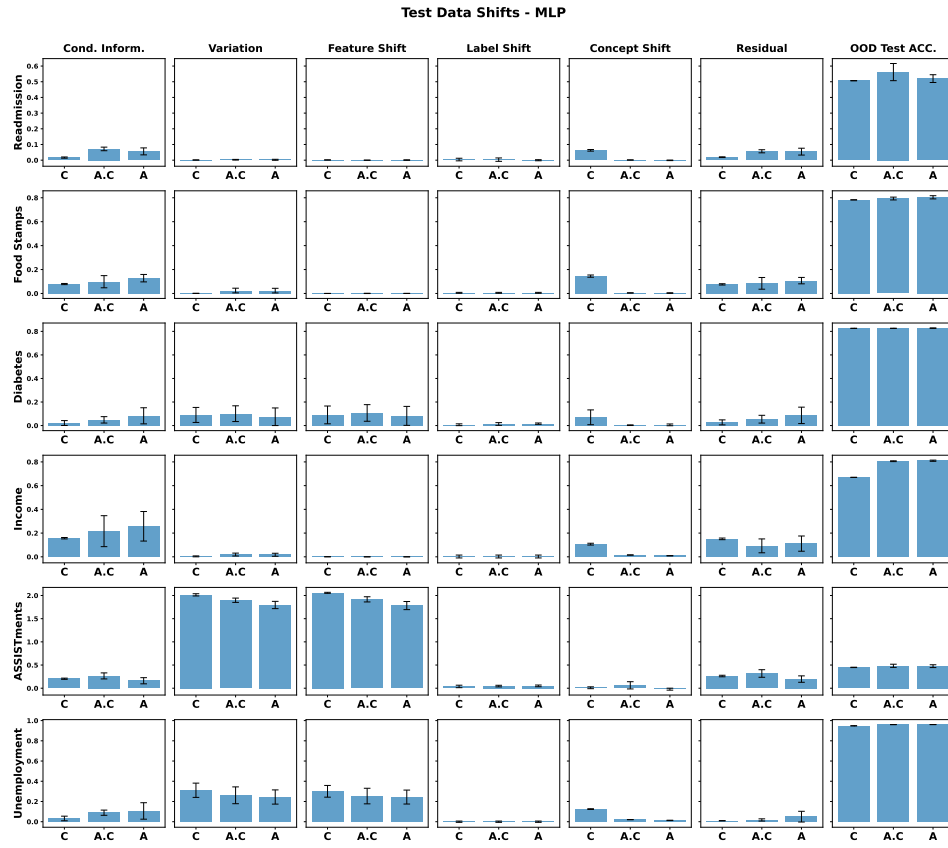


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

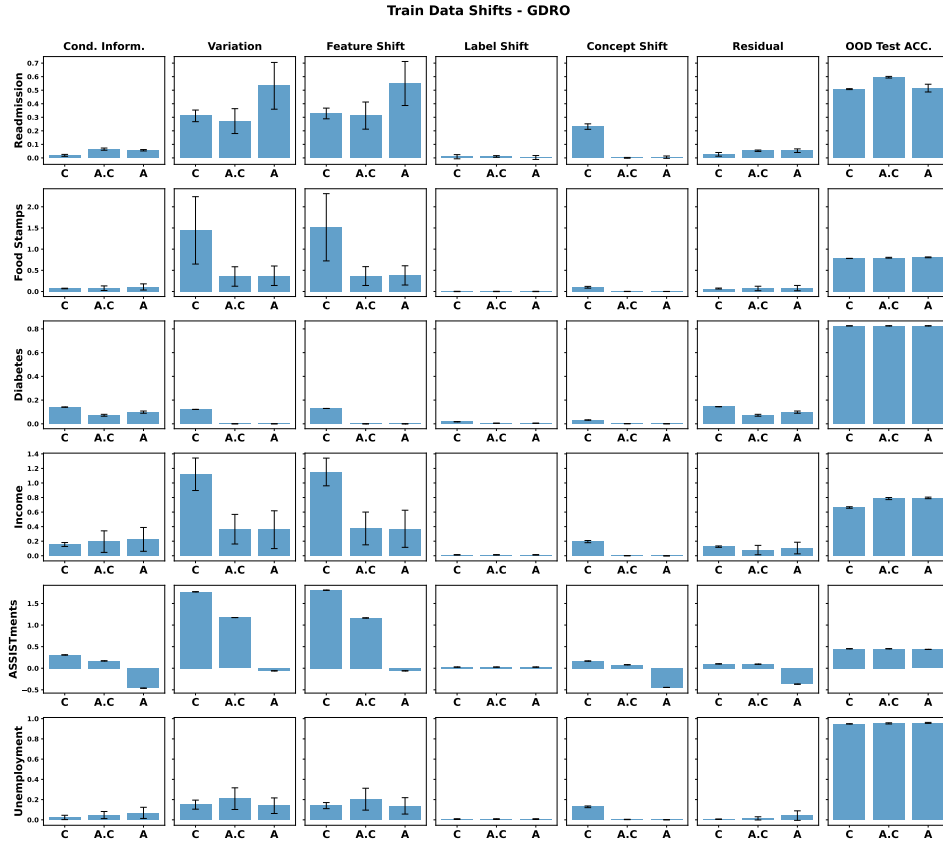


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

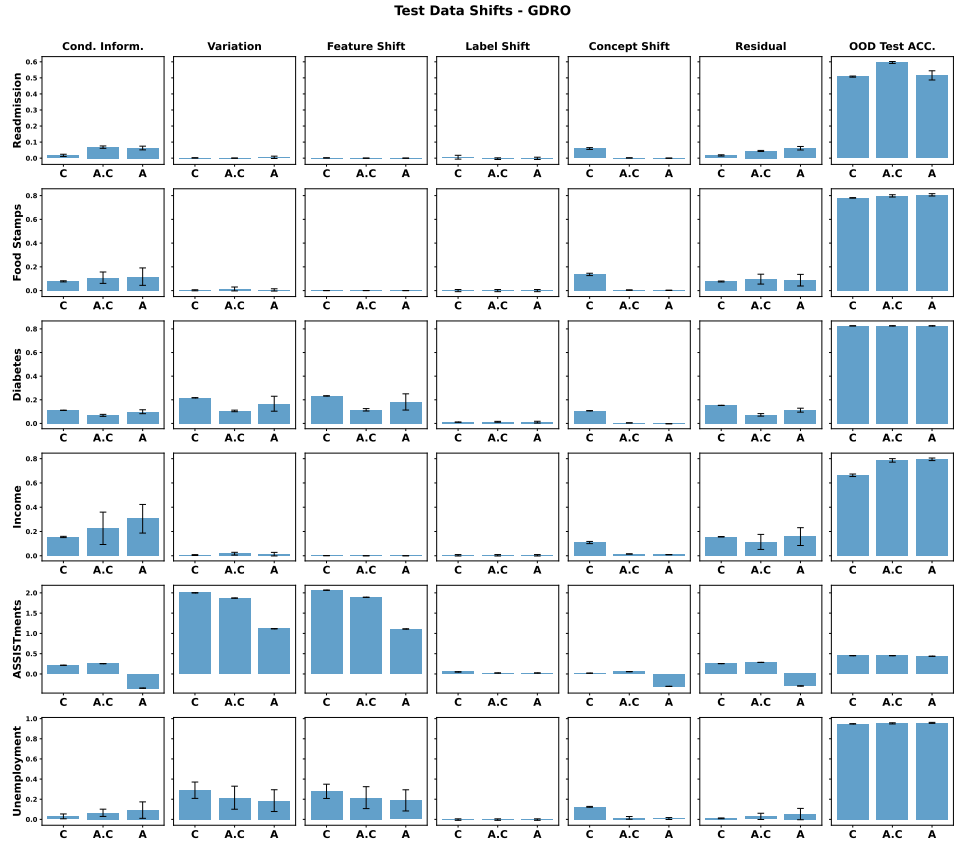


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

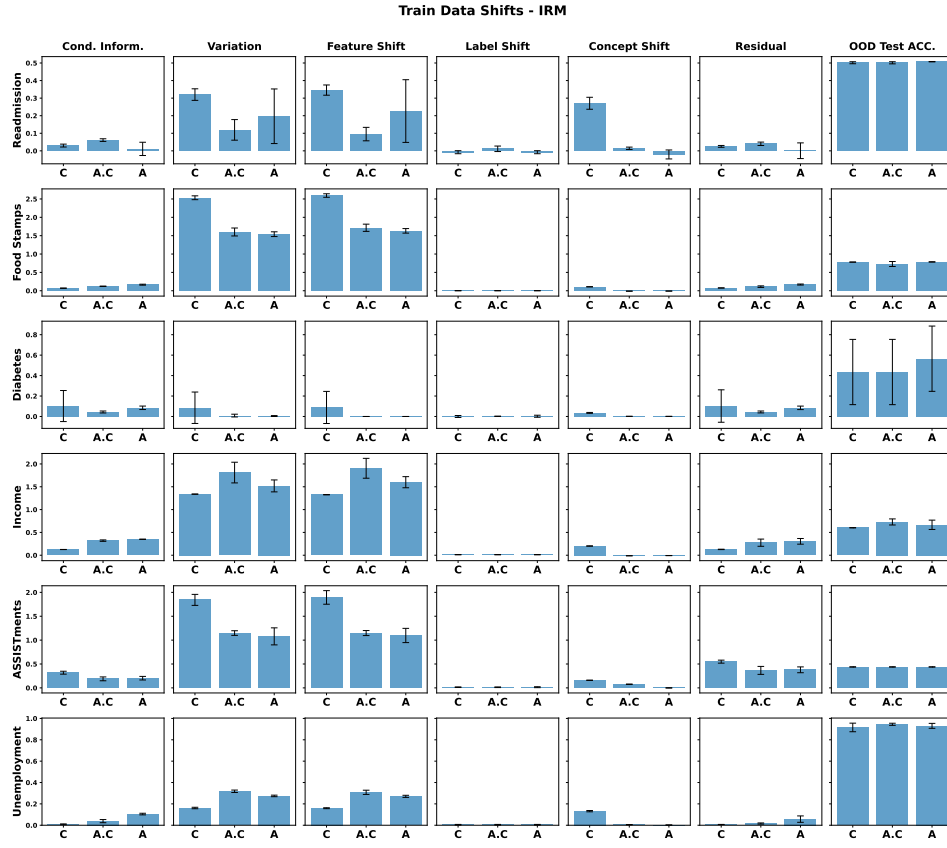


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

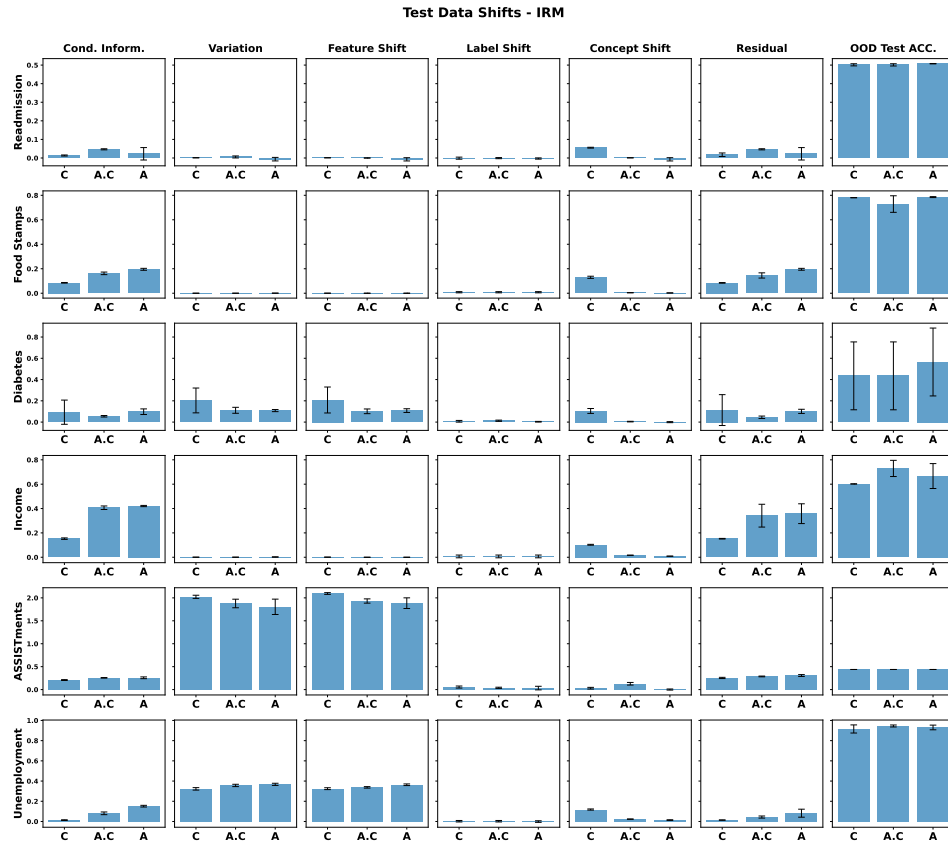


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

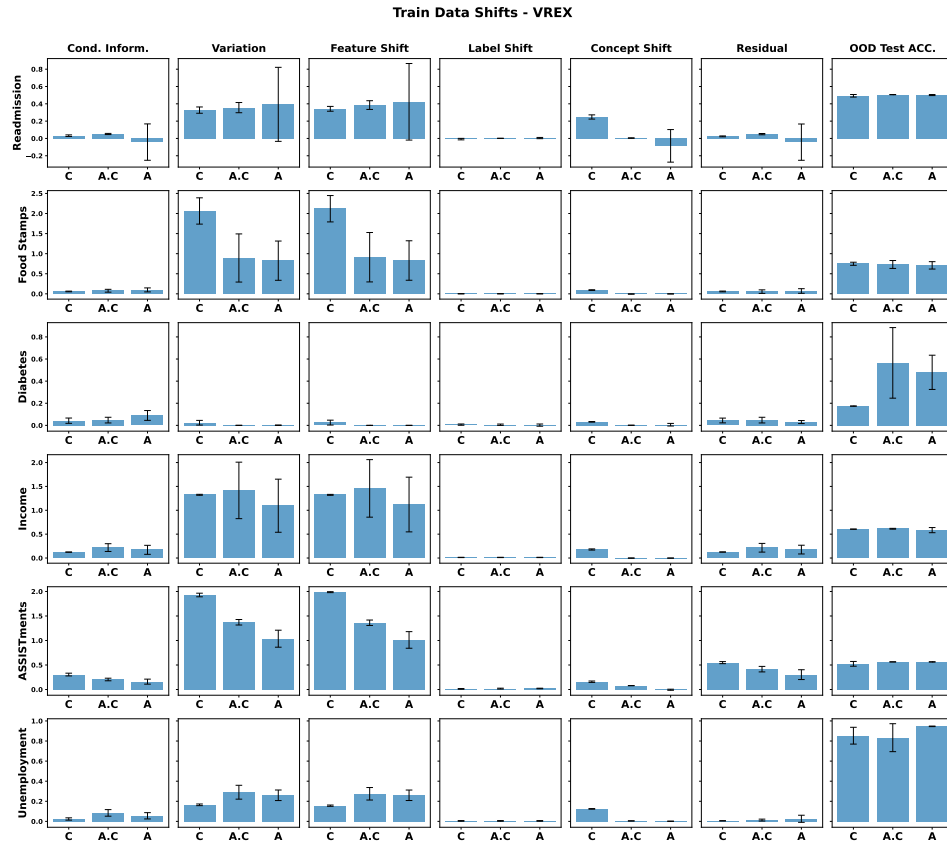


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

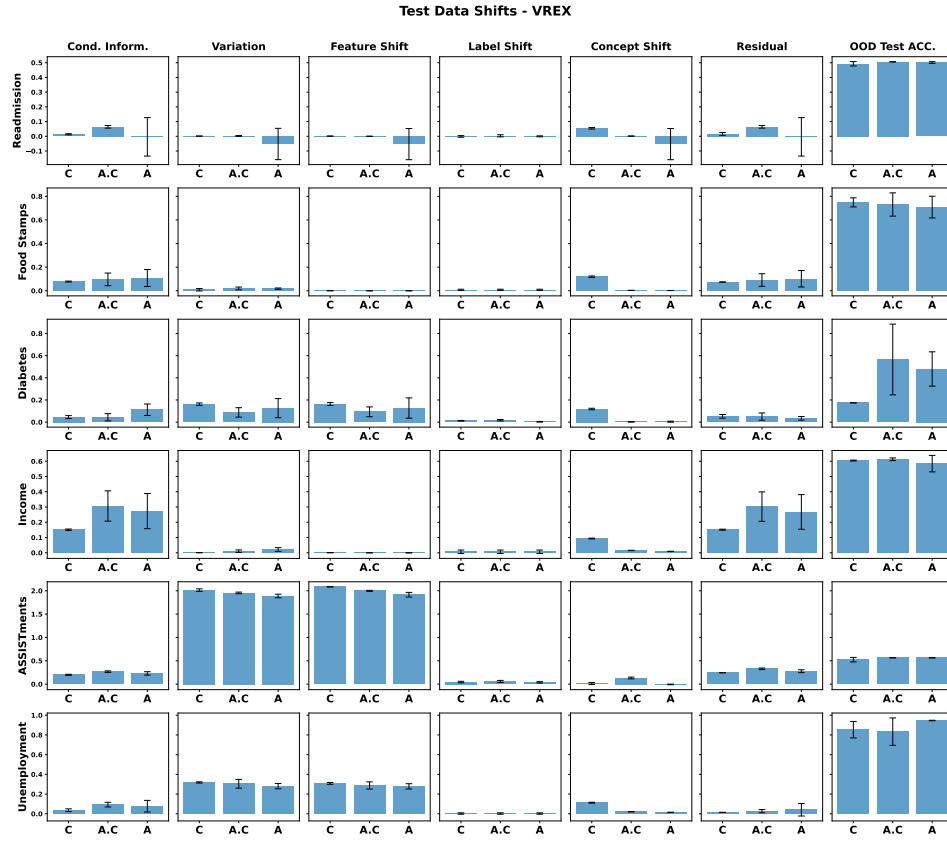


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

841 D Results on synthetic datasets

842 Because synthetic data lets us control every quantity exactly, we use it to isolate three effects that
 843 are hard to disentangle in the wild: (i) the impact of different informativeness criteria, (ii) how the
 844 degree of confounder overlap across environments affects performance, and (iii) how the ratio of
 845 causal to anti-causal variables shapes performance. The data-generating process follows the structural
 846 equations in (12), with causal structure $U \rightarrow X$, $U \rightarrow Y$, $U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is
 847 an unobserved confounder, X an observed covariate, Y the target, and $\mathbf{X}_I$ additional informative
 848 covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

849 We explore different functional forms for f_X , f_i , and f_Y , with the corresponding results presented in
 850 our analysis. Domain shifts are induced by systematically varying the environment-specific mean
 851 parameter μ_u^e of the confounder distribution. This allows us to examine how different degrees of
 852 distributional shift affect model performance across environments.

853 **Informativeness vs. accuracy:** We begin with linear structural equations $f_X = 0.3U$, $f_i = 0.1U$,
 854 $f_Y = X - 2U$, using $\mu_u^e \in \{-2, 2\}$ for training and $\mu_u^e \in \{0, 4\}$ for test environments—so training
 855 and test environments do not overlap, a deliberately demanding regime. In Figure 5 of the main paper
 856 (one hidden confounder, $|\mathbf{X}_I| = 20$), sweeping the number of informative covariates from 0 to 20
 857 moves all four quantities in the beneficial direction at once: MSE falls, conditional informativeness
 858 rises, feature shift rises, and concept shift falls. Two further sweeps confirm the effect is not specific
 859 to how informativeness is introduced. First, with a single informative covariate ($|\mathbf{X}_I| = 1$) and
 860 noise swept over $\sigma_i \in \{0, 0.1, \dots, 2.0\}$, Figure D16 shows that lowering noise from 2.0 to 0.0 again
 861 reduces MSE, raises conditional informativeness and feature shift, and lowers concept shift, with
 862 the variation term pinned at zero throughout. Second, scaling up to 20 hidden confounders with 20
 863 matched informative covariates (equations $f_X = 0.3U$, $f_i = 0.2U$, $f_Y = X - 2U$), Figure D17
 864 reproduces the identical four-way movement. Across all three sweeps the direction of every term
 865 is consistent, giving a controlled, monotonic demonstration that informativeness improves OOD
 866 generalization.

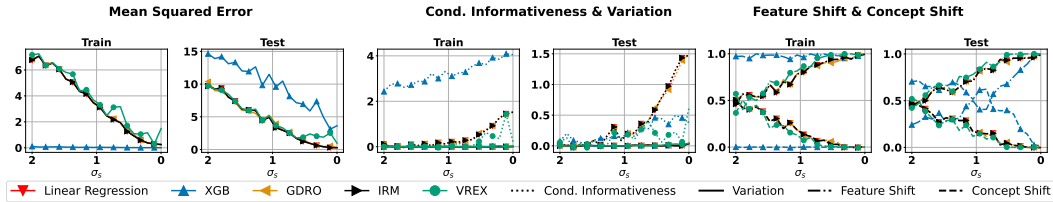


Figure D16: Single-proxy noise sweep ($|\mathbf{X}_I| = 1$, σ_i from 2.0 down to 0.0): a lower-noise proxy monotonically reduces MSE and concept shift while increasing conditional informativeness and feature shift; the variation term stays at zero.

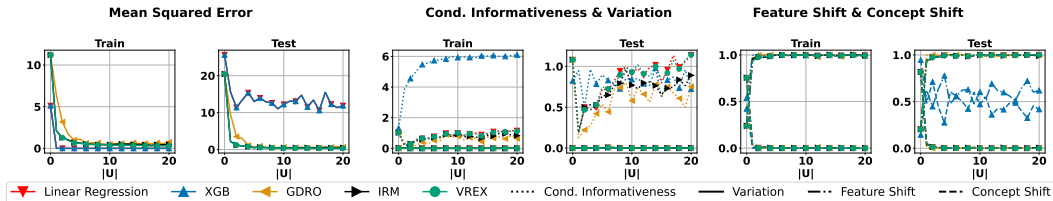


Figure D17: Multi-confounder setting (20 hidden confounders, 20 matched proxies): adding proxy variables $\mathbf{X}_I$ informative to Y again drives MSE and concept shift down while pushing conditional informativeness and feature shift up, reproducing the single-confounder trend at larger scale.

867 **Overlapping confounder support vs. accuracy:** The recent method of Prashant et al. [62] assumes
 868 test confounder support lies within training support. Having shown that correct relationships are

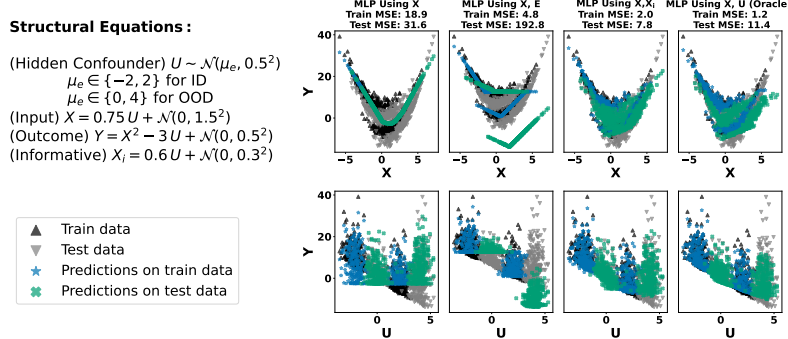


Figure D18: Performance of MLP on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

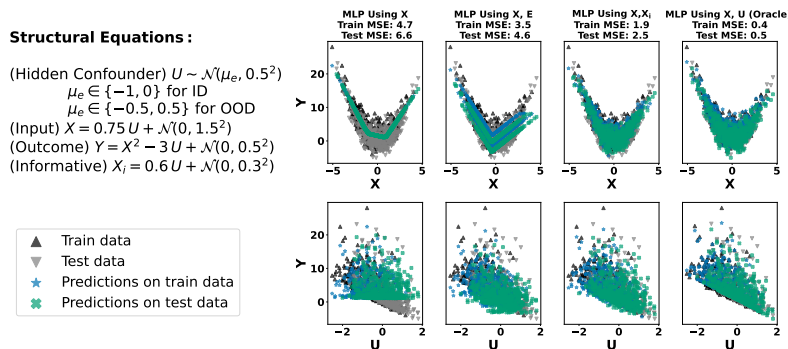


Figure D19: Performance of MLP on the high overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

recoverable in synthetic linear data when observed covariates carry enough information about the hidden confounder, we now stress-test this on nonlinear data with MLP and XGB under both low and high confounding overlap (Figures D18–D21). The comparison across feature sets is telling. In the low-overlap regime (distant μ_e between ID and OOD), a model using X alone learns an inadequate global function that cannot separate environments; adding environment-specific summary statistics of X recovers part of the gap; but the largest gains come only from adding informative covariates—the same ordering seen in the linear intro example. Oracle results and U - Y scatter plots are included to make the learned input–output mappings explicit. As expected, all methods do better under high overlap, but the more consequential finding is that informative covariates deliver clear improvements even in the low-overlap regime—a previously unstudied setting that directly challenges the common-support assumption in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] report that baseline methods (e.g., XGB, MLP) consistently match or beat dedicated OOD techniques (e.g., GDRO, IRM, VREX) on real-world tabular data—a pattern that coincides with the prevalent $\mathbf{X} \rightarrow Y$ structure. To test whether this baseline advantage is driven by that structure, we sweep the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines remain resilient across the entire range of ρ —including under the mixed structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ —achieving the higher PMA-OOD scores [24] (fraction of maximum OOD accuracy) in Figure D22, even though no single method dominates at every ρ (Figures D23, D24). This synthetic result mirrors the real-world ranking and thus generalizes the ERM-is-competitive finding beyond the benchmark datasets. The comparison spans 22 methods in three families: (1) ERM-based (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust optimization (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

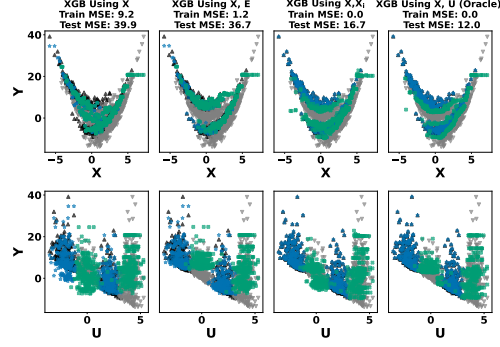


Figure D20: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

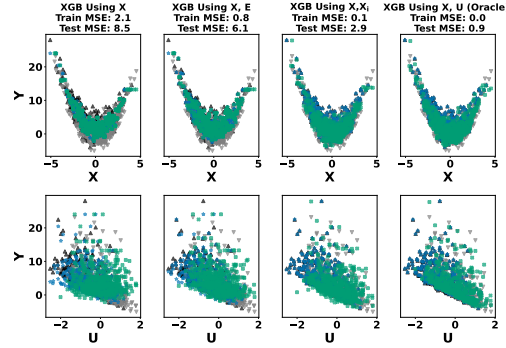


Figure D21: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_t|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

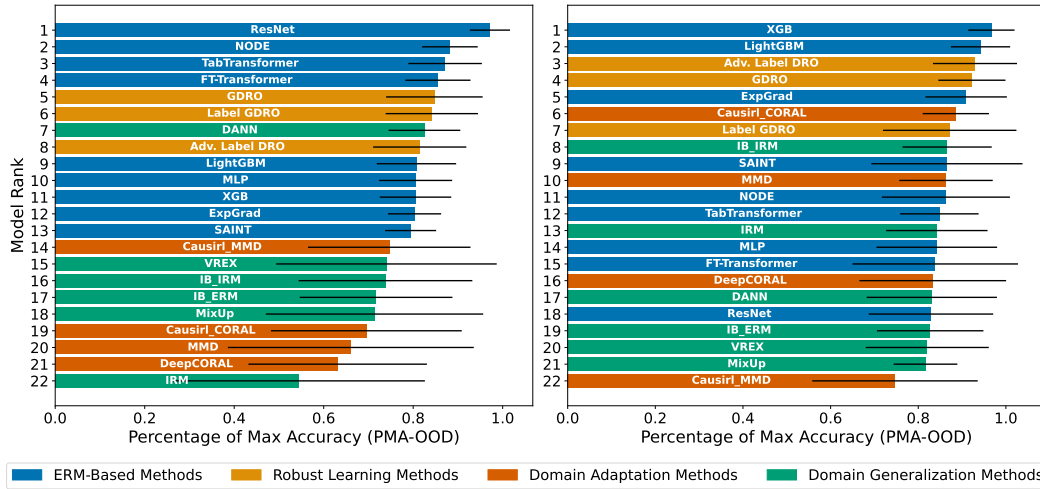


Figure D22: Across the causal/anti-causal sweep, ERM-based methods attain the highest percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, reproducing the real-world ranking reported by [24].

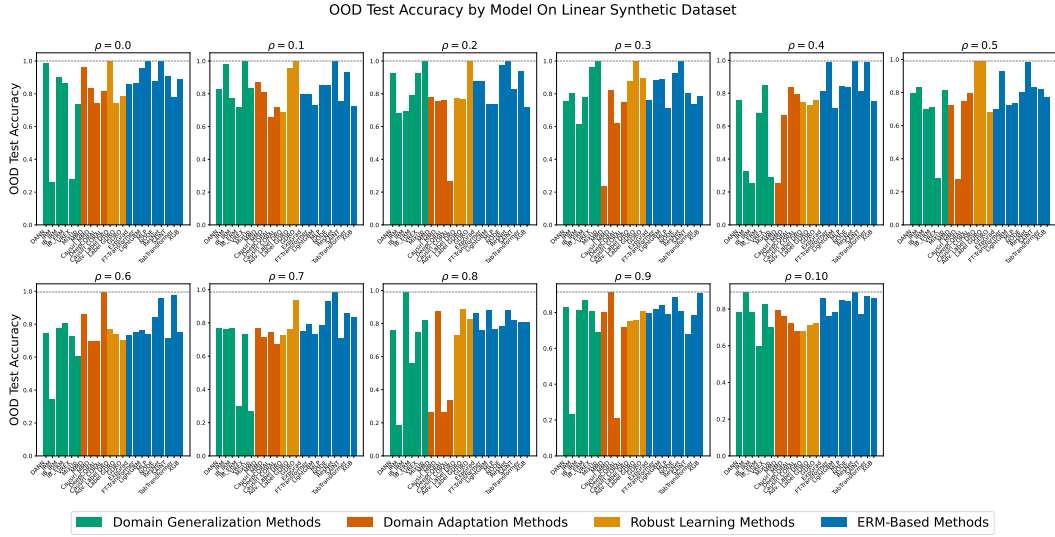


Figure D23: OOD test accuracy on the linear setting per model for different values of ρ .

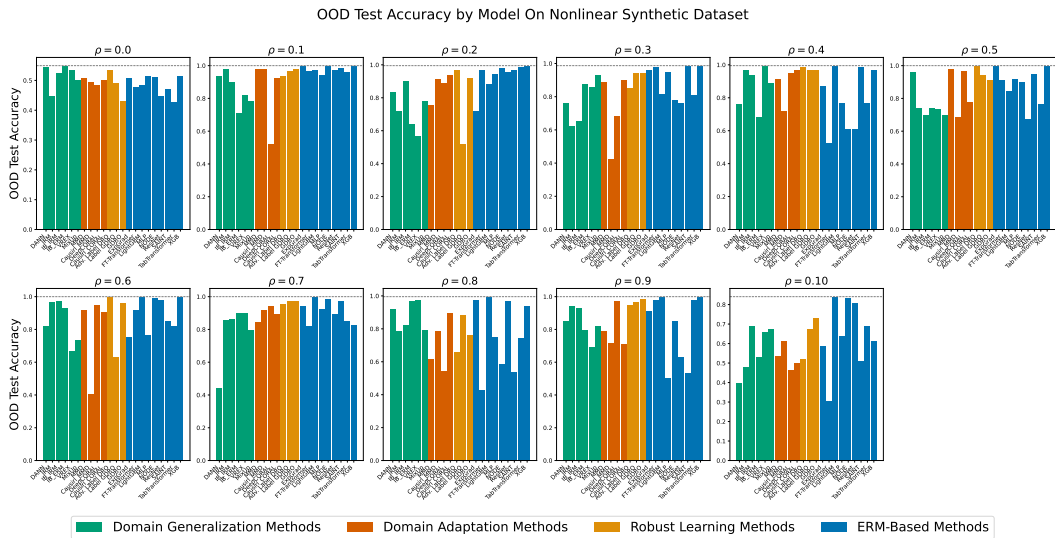


Figure D24: OOD test accuracy on the non-linear setting per model for different values of ρ

E Qualitative analysis of hidden confounding shift in real-world datasets

This section complements the quantitative evidence of Table 2 with a qualitative account of *which* hidden confounders plausibly drive the measured shifts. Tables E3-E18 enumerate candidate hidden confounders and the observed covariates they influence across a range of real-world datasets [57] (see [24, 57] for dataset details). We obtain the candidates via GPT4o [1] and note the lists are illustrative rather than exhaustive. Read alongside the significant-on-every-dataset shift measurements, this qualitative evidence reinforces the case that hidden confounding shift is both present and consequential across domains.

Prompt: We use the prompt below to query GPT4o to get possible hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.